

Photo Style Transfer (?) (DPST) after our synthesized image. As shown in Fig 13, compared to the direct synthesized image at row 2, the image after DPST possess a higher color consistency while preserving the pathological lesions such as microaneurysms.

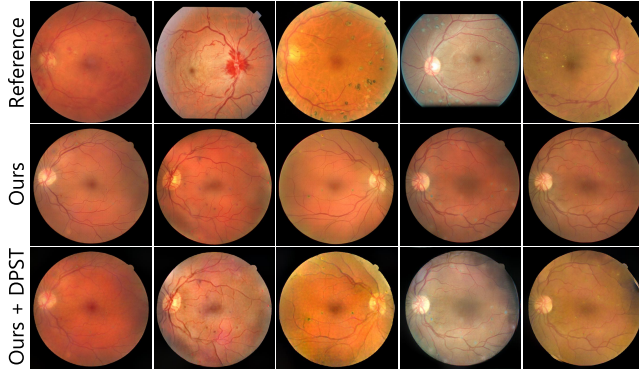


Figure 13: Color transfer results. Our method with a post process achieves a better visual effect.

Computation Time

All the experiments are tested out on a sever with Intel Xeon E5-2643 CPU, 256GB memory and Titan-Xp GPU. Training time on DRIVE and testing time are shown in table 2. Compared to Tub-sGAN (?), we are faster which benefits from a more streamlined feature extraction network and descriptor-based comparison.

	Training	Testing	Platform
Tub-sGAN	108/109 min	0.45/0.12 s	Titan-X/Xp
Ours	90 min	0.12 s	Titan-Xp

Table 2: Computation time of the different methods. The Tub-sGAN time on Titan-X is reported in their paper, and its time on Titan-Xp is measured by us.

Conclusion

To exploit the network interpretability in medical imaging, we proposed a novel strategy to encode the descriptor from the activated neurons that directly related to the prediction. To visually illustrate the extracted pathologic descriptor, we followed the similar methodology of Koch’s Postulates that aim to identify the unknown pathogen. In addition, we proposed a GAN based visualization method to visualize the pathological descriptor into a fully controllable pathology retinal image from an unseen binary vessel segmentation. The retinal images we generated have shown medical plausible symptoms as the reference image. Since pathological descriptor is associated with individual lesion and spatial independent, we could arbitrarily manipulate the position and quantity of the symptom. We verified the generated images with a group of licensed ophthalmologists and our result

is shown to be both qualitatively and quantitatively superior to state-of-the-art. The feedback of doctors shows our strategy has strengthened their understanding on how deep learning makes prediction. Not limited in interpreting medical imaging, we will extend our strategy to more general interpretability problem.

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